

DEPARTMENT OF MATHEMATICS
MAT122 INTRODUCTORY MATHEMATICS II
WEEK 5: LECTURE NOTE

DIFFERENTIATION continued....

Differentiation of logarithmic and exponential functions

Differentiation of Logarithmic Functions

Consider the function

$$f(x) = (1 + x)^{\frac{1}{x}}.$$

Numerical evaluation of the values of $f(x)$ as x approaches 0 leads us to conclude that

$$\lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}} = e \approx 2.71828182846.$$

Recall that the function

$$y = \ln x = \log_e x$$

is called the **NATURAL LOGARITHMIC FUNCTION**.

Theorem 1 (Derivative of the Natural Logarithmic Function). *Let u be a differentiable function of x .*

$$(1) \quad \frac{d}{dx}[\ln x] = \frac{1}{x}, \quad x > 0 \qquad (2) \quad \frac{d}{dx}[\ln u] = \frac{1}{u} \frac{du}{dx} = \frac{u'}{u}, \quad u > 0$$

Example 2. *Differentiation of Logarithmic Functions*

$$\begin{aligned} 1. \quad & \frac{d}{dx}[\ln(2x)] = \frac{1}{2x} \frac{d}{dx}[2x] = \frac{1}{x}, \quad x > 0 \\ 2. \quad & \frac{d}{dx}[\ln(x^2 + 1)] = \frac{1}{x^2 + 1} \frac{d}{dx}[x^2 + 1] = \frac{2x}{x^2 + 1} \end{aligned}$$

Example 3 (Logarithmic Properties as Aids to Differentiation). *Differentiate*

$$f(x) = \ln \sqrt{x+1}.$$

Solution. *Because*

$$f(x) = \ln \sqrt{x+1} = \frac{1}{2} \ln(x+1)$$

we can write

$$f'(x) = \frac{1}{2} \left(\frac{1}{x+1} \right) = \frac{1}{2(x+1)}, \quad x > -1$$

Example 4 (Logarithmic Differentiation). *Find the derivative of*

$$y = \frac{(x-2)^2}{\sqrt{x^2+1}}, \quad x \neq 2.$$

Solution. Note that $y > 0$ for all $x \neq 2$. So, $\ln y$ is defined. Begin by taking the natural logarithm of each side of the equation. Then apply logarithmic properties and differentiate implicitly. Finally, solve for y' .

$y = \frac{(x-2)^2}{\sqrt{x^2+1}}, \quad x \neq 2$	Write original equation.
$\ln y = \ln \frac{(x-2)^2}{\sqrt{x^2+1}}$	Take natural log of each side.
$\ln y = 2 \ln(x-2) - \frac{1}{2} \ln(x^2+1)$	Logarithmic properties
$\frac{y'}{y} = 2 \frac{1}{x-2} - \frac{1}{2} \left(\frac{2x}{x^2+1} \right)$	Differentiate.
$\frac{y'}{y} = \frac{x^2+2x+2}{(x-2)(x^2+1)}$	Simplify.
$y' = y \frac{x^2+2x+2}{(x-2)(x^2+1)}$	Solve for y'
$y' = \frac{(x-2)^2}{\sqrt{x^2+1}} \left[\frac{x^2+2x+2}{(x-2)(x^2+1)} \right]$	Substitute for y
$y' = \frac{(x-2)(x^2+2x+2)}{(x^2+1)^{\frac{3}{2}}}$	Simplify.

Differentiation of Exponential Functions

Definition 5 (Definition of the Natural Exponential Function). *The inverse function of the natural logarithmic function $f(x) = \ln x$ is called the natural exponential function and is denoted by*

$$f^{-1}(x) = e^x.$$

That is,

$$y = e^x \quad \text{if and only if} \quad x = \ln y.$$

The inverse relationship between the natural logarithmic function and the natural exponential function can be summarized as shown.

$$\ln e^x = x \quad \text{and} \quad e^{\ln x} = x. \quad \text{Inverse relationship}$$

Example 6. Solve

$$(a) \quad 7 = e^{x+1} \qquad (b) \quad \ln(2x-3) = 5$$

Solution.

(a) We can convert from exponential form to logarithmic form by taking the natural logarithm of each side of the equation.

$$7 = e^{x+1} \Rightarrow \ln 7 = \ln e^{x+1} \Rightarrow \ln 7 = x+1 \Rightarrow x = \ln 7 - 1.$$

Check!!

$$e^{(\ln 7 - 1) + 1} = e^{\ln 7} = 7 \quad \checkmark$$

(b) To convert from logarithmic form to exponential form, you can exponentiate each side of the logarithmic equation.

$$\ln(2x - 3) = 5 \Rightarrow e^{\ln(2x-3)} = e^5 \Rightarrow 2x - 3 = e^5 \Rightarrow x = \frac{e^5 + 3}{2}.$$

Check!!

$$\ln[2(\frac{e^5 + 3}{2}) - 3] = \ln(e^5 + 3 - 3) = \ln e^5 = 5 \quad \checkmark$$

Theorem 7. Let u be a differentiable function of x .

$$(1) \quad \frac{d}{dx}[e^x] = e^x \qquad (2) \quad \frac{d}{dx}[e^u] = e^u \frac{du}{dx} = e^u u'$$

Definition 8. If a is a positive real number ($a \neq 1$) and x is any real number, then the **exponential function to the base a** is denoted by a^x and is defined by

$$a^x = e^{(\ln a)x}.$$

If $a = 1$, then $y = 1^x = 1$ is a constant function.

Definition 9 (Definition of Logarithmic Function to Base a). If a is a positive real number ($a \neq 1$) and x is any positive real number, then the **logarithmic function to the base a** is denoted by $\log_a x$ and is defined by

$$\log_a x = \frac{1}{\ln a} \ln x.$$

Theorem 10 (Derivatives for Bases Other than e). Let a be a positive real number ($a \neq 1$) and let u be a differentiable function of x .

$$\begin{aligned} 1. \quad & \frac{d}{dx}[a^x] = (\ln a)a^x \\ 2. \quad & \frac{d}{dx}[a^u] = (\ln a)a^u \frac{du}{dx} = (\ln a)a^u u' \\ 3. \quad & \frac{d}{dx}[\log_a x] = \frac{1}{(\ln a)x} \\ 4. \quad & \frac{d}{dx}[\log_a u] = \frac{1}{(\ln a)u} \frac{du}{dx} = \frac{u'}{(\ln a)u} \end{aligned}$$

Example 11. Find the derivative of each function.

$$(a) \quad y = 2^x \qquad (b) \quad y = 2^{3x} \qquad (c) \quad y = \log_{10}(\cos x) \qquad (d) \quad y = \log_3 \frac{\sqrt{x}}{x+5}$$

Solution.

$$(a) \quad y' = \frac{d}{dx}[2^x] = (\ln 2)2^x.$$

$$(b) \quad y' = \frac{d}{dx}[2^{3x}] = (\ln 2)2^{3x} \frac{d}{dx}[3x] = (3 \ln 2)2^{3x}.$$

$$(c) \quad y' = \frac{d}{dx}[\log_{10}(\cos x)] = \frac{1}{(\ln 10)\cos x} \frac{d}{dx}[\cos x] = \frac{-\sin x}{(\ln 10)\cos x} = -\frac{1}{\ln 10} \tan x.$$

(d) Note that

$$y = \log_3 \frac{\sqrt{x}}{x+5} = \frac{1}{2} \log_3 x - \log_3 (x+5).$$

Therefore,

$$\begin{aligned} y' &= \frac{d}{dx} [\log_3 \frac{\sqrt{x}}{x+5}] \\ &= \frac{1}{2} \frac{d}{dx} [\log_3 x] - \frac{d}{dx} [\log_3 (x+5)] \\ &= \frac{1}{(2 \ln 3)x} - \frac{1}{(\ln 3)(x+5)} = \frac{5-x}{(\ln 9)x(x+5)} \end{aligned}$$

Theorem 12 (The Power Rule for Real Exponents). *Let n be any number, and let u be a differentiable function of x .*

$$\begin{aligned} (1) \quad & \frac{d}{dx} [x^n] = nx^{n-1} \\ (2) \quad & \frac{d}{dx} [u^n] = nu^{n-1} \frac{du}{dx} = nu^{n-1} u'. \end{aligned}$$

Example 13. Comparing Variables and Constants

(a) $\frac{d}{dx} [e^e] = 0$	Constant Rule
(b) $\frac{d}{dx} [e^x] = e^x$	Exponential Rule
(c) $\frac{d}{dx} [x^e] = ex^{e-1}$	Power Rule

Implicit Differentiation

Up to this point, most functions have been expressed in explicit form. For example, in the equation $y = 3x^2 - 5$, the variable y is explicitly written as a function of x . Some functions, however, are only implied by an equation. For instance, the function $y = \frac{1}{x}$ is defined implicitly by the equation

$$xy = 1. \qquad \text{IMPLICIT FORM}$$

To find $\frac{dy}{dx}$ for this equation, you can write y explicitly as a function of x and then differentiate

<i>Implicit Form</i>	<i>Explicit Form</i>	<i>Derivative</i>
$xy = 1$	$y = \frac{1}{x} = x^{-1}$	$\frac{dy}{dx} = -x^{-2} = -\frac{1}{x^2}$

This strategy works whenever you can solve for the **function explicitly**. You cannot, however, use this procedure when you are unable to solve for y as a function of x . For instance, how would you find $\frac{dy}{dx}$ for the equation

$$x^2 - 2y^3 + 4y = 2?$$

For this equation, it is difficult to express y as a function of x explicitly. To find $\frac{dy}{dx}$ you can use **implicit differentiation**.

To understand how to find $\frac{dy}{dx}$ implicitly, you must realize that the differentiation is taking place with respect to x . This means that when you differentiate terms involving x alone, you differentiate as usual. However, when you differentiate terms involving y , you must apply Chain Rule, because, you are assuming that y is defined implicitly as a differentiable function of x .

Example 14. Differentiating with Respect to x

$$1. \frac{d}{dx}[x^3] = 3x^2 \quad \text{Variables agree: use Simple Power Rule}$$

$$2. \frac{d}{dx}[y^3] = 3y^2 \frac{dy}{dx} \quad \text{Variables disagree: use Chain Rule}$$

$$3. \frac{d}{dx}[x + 3y] = \frac{d}{dx}[x] + \frac{d}{dx}[3y] = 1 + 3 \frac{dy}{dx}$$

$$4. \frac{d}{dx}[xy^2] = y^2 \frac{d}{dx}[x] + x \frac{d}{dx}[y^2] = y^2 + 2xy \frac{dy}{dx} \quad \text{Use of Product Rule and Chain Rule}$$

GUIDELINES FOR IMPLICIT DIFFERENTIATION

1. Differentiate both sides of the equation with respect to x .
2. Collect all terms involving $\frac{dy}{dx}$ on the left side of the equation and move all other terms to the right side of the equation.
3. Factor $\frac{dy}{dx}$ out of the left side of the equation.
4. Solve for $\frac{dy}{dx}$.

Example 15 (Implicit Differentiation). Find $\frac{dy}{dx}$ given that

$$y^3 + y^2 - 5y - x^3 = -4.$$

Solution.

1. Differentiate both sides of the equation with respect to x .

$$\begin{aligned} \frac{d}{dx}[y^3 + y^2 - 5y - x^3] &= \frac{d}{dx}[-4] \\ \frac{d}{dx}[y^3] + \frac{d}{dx}[y^2] - \frac{d}{dx}[5y] - \frac{d}{dx}[x^3] &= \frac{d}{dx}[-4] \\ 3y^2 \frac{dy}{dx} + 2y \frac{dy}{dx} - 5 \frac{dy}{dx} - 2x &= 0 \end{aligned}$$

2. Collect the $\frac{dy}{dx}$ terms on the left side of the equation and move all other terms to the right side of the equation.

$$3y^2 \frac{dy}{dx} + 2y \frac{dy}{dx} - 5 \frac{dy}{dx} = 2x.$$

3. Factor $\frac{dy}{dx}$ out of the left side of the equation.

$$(3y^2 + 2y - 5)\frac{dy}{dx} = 2x.$$

4. Solve for $\frac{dy}{dx}$ by dividing by $(3y^2 + 2y - 5)$.

$$\frac{dy}{dx} = \frac{2x}{3y^2 + 2y - 5}$$

Example 16 (Finding the Slope of a Graph Implicitly). Determine the slope of the tangent line to the graph of $x^2 + 4y^2 = 4$ at the point $(\sqrt{2}, -\frac{1}{\sqrt{2}})$.

Solution. Differentiating the equation with respect to x , we get

$$2x + 8y\frac{dy}{dx} = 0.$$

Solving for $\frac{dy}{dx}$ and simplifying, we have

$$\frac{dy}{dx} = -\frac{x}{4y}.$$

The slope of the tangent line at the point $(\sqrt{2}, -\frac{1}{\sqrt{2}})$ is obtained by evaluating $\frac{dy}{dx}$ when $x = \sqrt{2}$ and $y = -\frac{1}{\sqrt{2}}$. So that the slope is given by

$$\frac{dy}{dx}(\sqrt{2}, -\frac{1}{\sqrt{2}}) = -\frac{\sqrt{2}}{-\frac{1}{\sqrt{2}}} = 2.$$

Example 17 (Determining a Differentiable Function). Find $\frac{dy}{dx}$ implicitly for the equation

$$\sin y = x.$$

Then find the largest interval of the form $-a < y < a$ on which y is a differentiable function of x .

Solution Note that

$$\begin{aligned}\frac{d}{dx}[\sin y] &= \frac{d}{dx}[x] \\ \cos y \frac{dy}{dx} &= 1 \\ \boxed{\frac{dy}{dx} = \frac{1}{\cos y}}.\end{aligned}$$

Therefore, the largest interval about the origin for which y is a differentiable function of x is $-\frac{\pi}{2} < y < \frac{\pi}{2}$. To see this, note that $\cos y$ is positive for all y in this interval and is 0 at

the endpoints. When you restrict to the interval you should be able to write explicitly as a function of x . To do this, you can use

$$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$$

and conclude that

$$\boxed{\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}}}.$$

Example 18 (Finding the Second Derivative Implicitly). Find $\frac{d^2y}{dx^2}$ implicitly for the equation

$$x^2 + y^2 = 25.$$

Solution. Differentiating each term with respect to x produces

$$\begin{aligned} 2x + 2y \frac{dy}{dx} &= 0 \\ \frac{dy}{dx} &= -\frac{x}{y}. \end{aligned}$$

Differentiating a second time with respect to x yields

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{d}{dx} \left[\frac{dy}{dx} \right] = -\frac{d}{dx} \left[\frac{x}{y} \right] \\ &= -\frac{y \frac{d}{dx}[x] - x \frac{dy}{dx}}{y^2} && \text{Quotient Rule} \\ &= -\frac{y - x \frac{dy}{dx}}{y^2} \\ &= -\frac{y - x(-\frac{x}{y})}{y^2} && \text{Substitute } -\frac{x}{y} \text{ for } \frac{dy}{dx} \\ &= -\frac{y^2 + x^2}{y^3} && \text{Simplify} \\ &= -\frac{25}{y^3} && \text{Substitute } 25 \text{ for } x^2 + y^2. \end{aligned}$$

Example 19 (Finding a Tangent Line to a Graph). Find the tangent line to the graph of $x^2(x^2 + y^2) = y^2$ at the point $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$.

Solution. By rewriting and differentiating implicitly, we obtain

$$\frac{dy}{dx} = \frac{x(2x^2 + y^2)}{y(1 - x^2)}.$$

Therefore, the slope is

$$\frac{dy}{dx} \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right) = 3$$

and the equation of the tangent line at this point is

$$y - \frac{\sqrt{2}}{2} = 3 \left(x - \frac{\sqrt{2}}{2} \right)$$

$$\boxed{y = 3x - \sqrt{2}}$$